

QML grid results: coverage and depth requirements for a clean B-mode null test

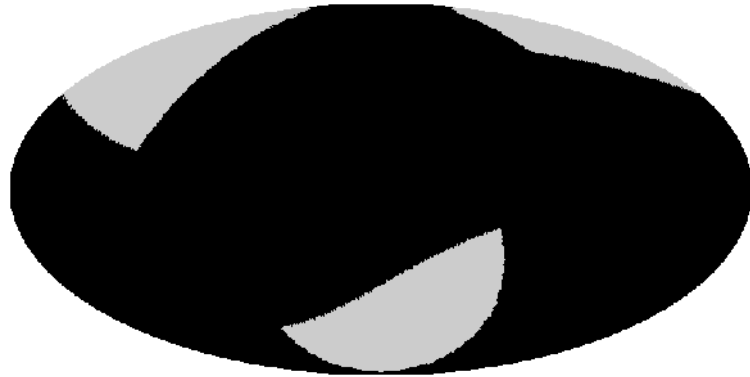
This report summarises a 3x3 grid of QML / PCI / heracles power-spectrum estimates for the GLASS full-sky weak-lensing simulation, varying the galaxy density ($n = 1, 10, 20$ /arcmin², rows in the grids below) and the galactic+ecliptic footprint cut (55-125, 65-115, 75-105 deg colatitude, columns below - sky coverage increases left to right). There is no true cosmological B-mode in this simulation, so a well-behaved measurement should recover BB power consistent with pure shot noise: the "BB/noise" column below is the measured QML $|C_{\ell}^{BB}|$ divided by the theory noise level (around 1 means the B-mode is fully explained by noise, i.e. "reduced").

Main finding: seeing a convincingly reduced B-mode requires BOTH good sky coverage AND good galaxy density.

- Coverage controls leakage. With the narrowest footprint (55-125 deg, the most masked sky), the recovered BB power exceeds the noise floor by 1-2 orders of magnitude at every density - mask-induced E-to-B leakage overwhelms the null signal regardless of how many galaxies are used.
- Density controls precision. With the widest footprint (75-105 deg), BB stays close to the noise floor at every density, but the noise floor itself only becomes small (a tight, informative null test) once the density is high ($n = 20$ /arcmin²); at $n = 1$ /arcmin² the same "consistent with noise" result carries far larger error bars and is much less informative.
- Only good coverage + good density together give both a low absolute B-mode amplitude and a tight measurement of it. The intermediate footprint (65-115 deg) shows this trade-off clearly: consistent with noise at low density, but with growing low- ℓ excess as density increases, since noise shrinks faster than the mask's residual leakage.

n [/arcmin ²]	footprint [deg]	BB/noise ($\ell < 30$)	BB/noise (all ℓ)
1	(55, 125)	782.83	566.82
1	(65, 115)	1.22	1.01
1	(75, 105)	0.89	0.94
10	(55, 125)	148.39	79.04
10	(65, 115)	1.76	1.36
10	(75, 105)	1.14	1.05
20	(55, 125)	153.28	134.08
20	(65, 115)	4.66	2.69
20	(75, 105)	1.25	1.13

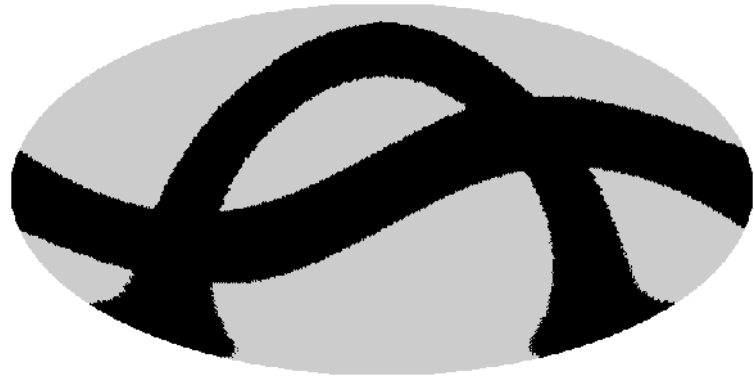
(55, 125) deg



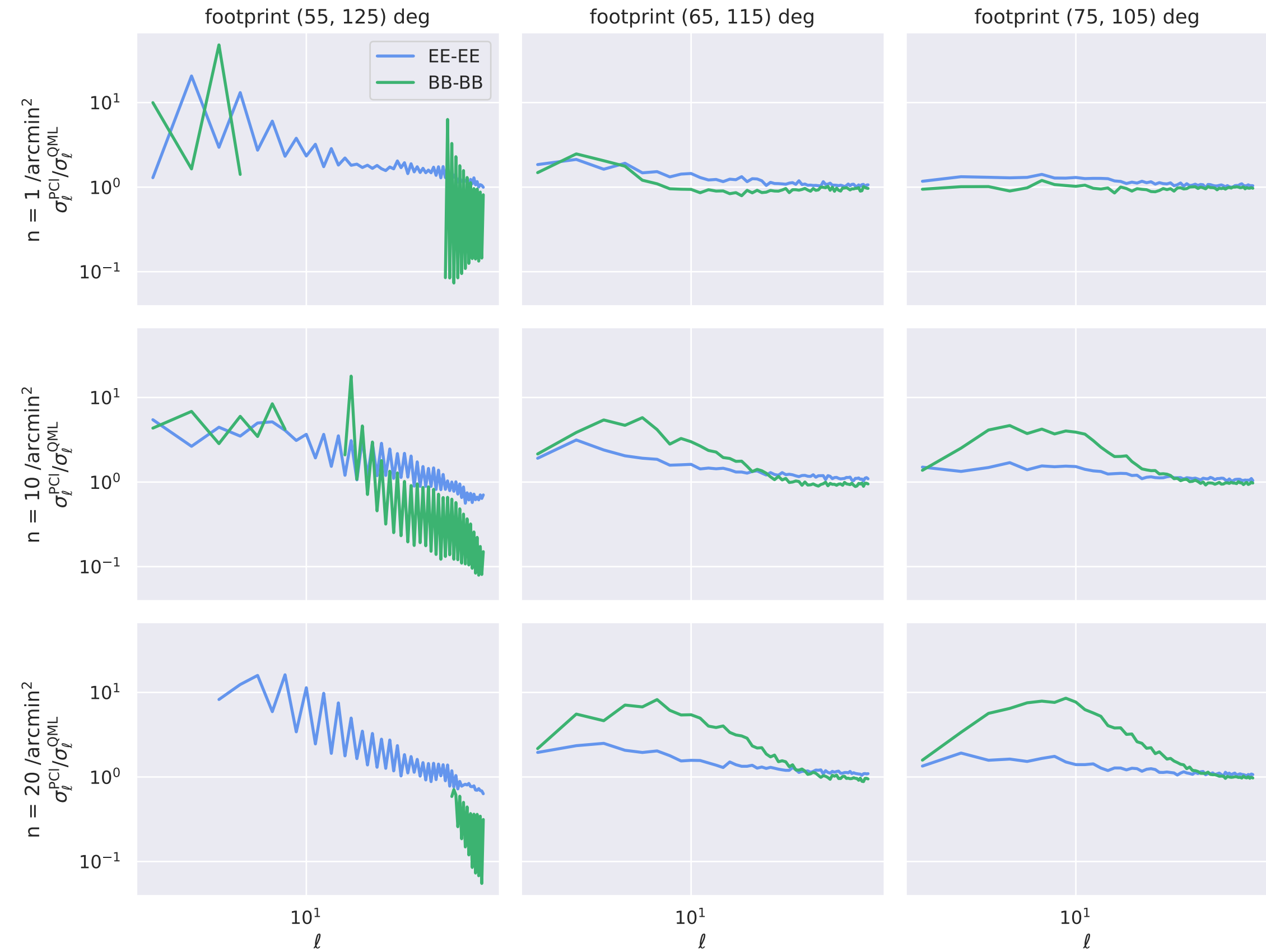
(65, 115) deg



(75, 105) deg

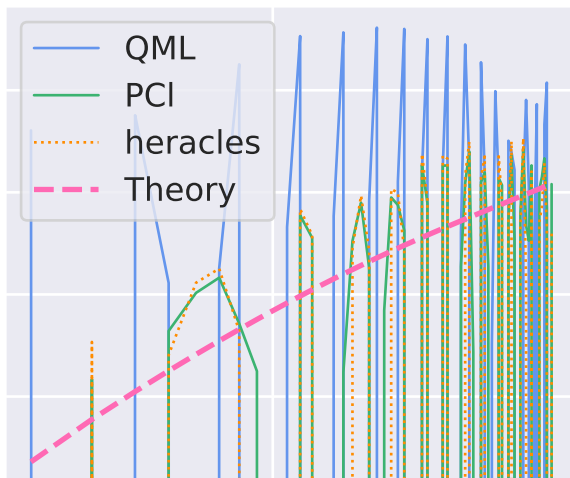


PCI / QML covariance ratio

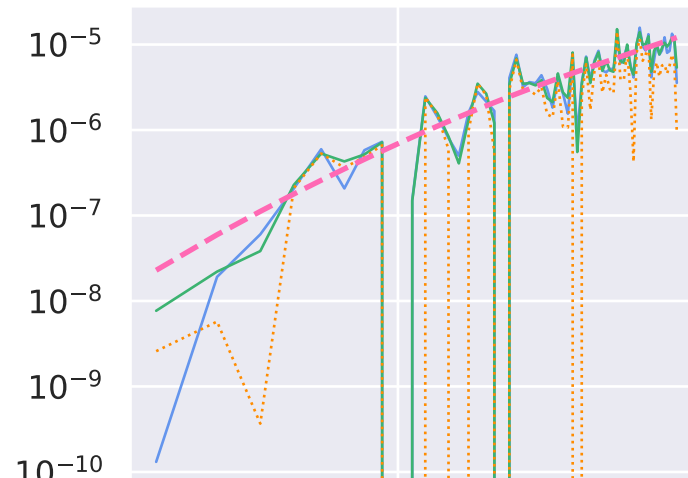


EE power spectrum estimates

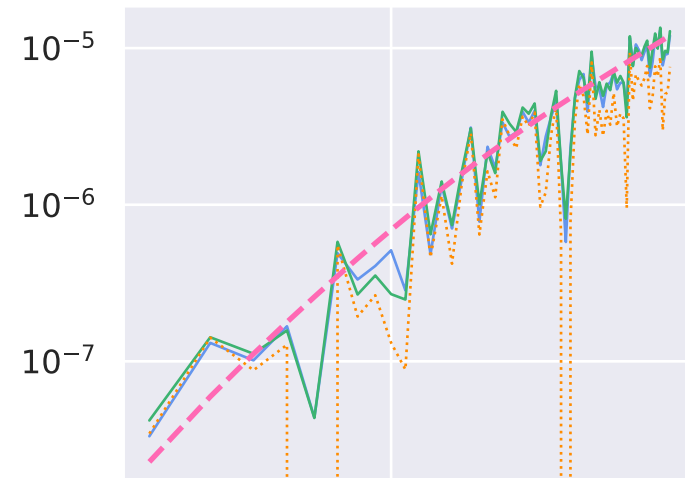
footprint (55, 125) deg



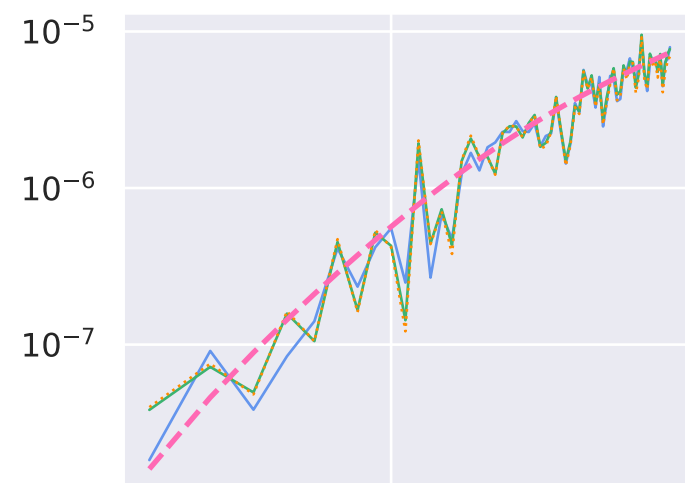
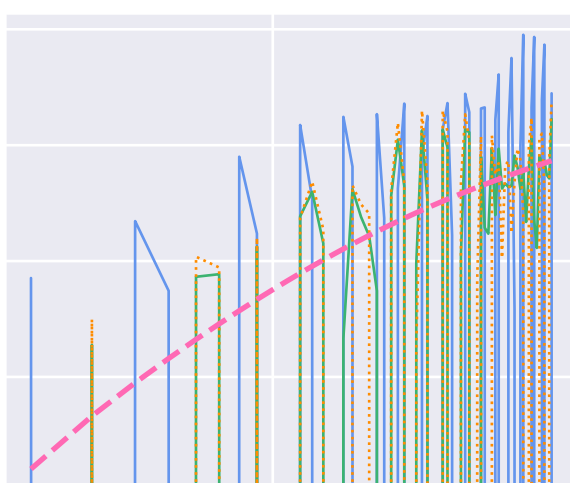
footprint (65, 115) deg



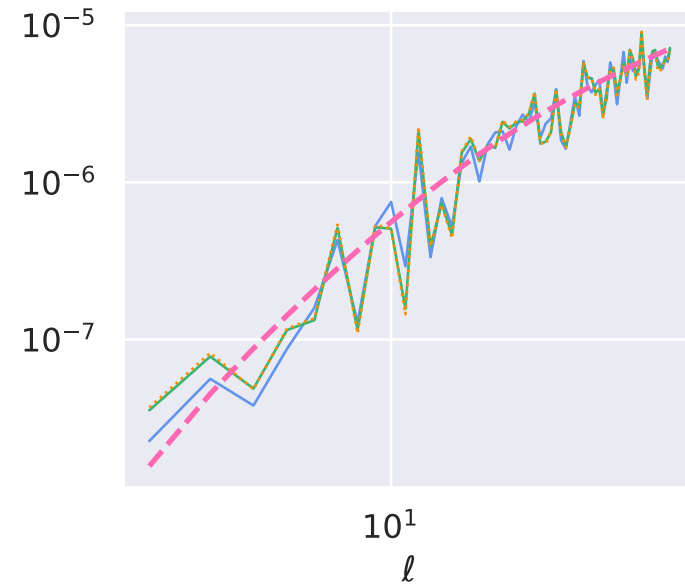
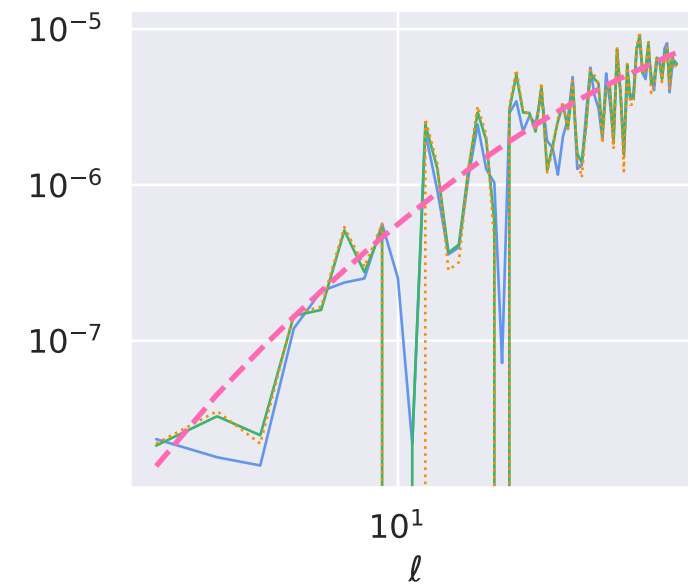
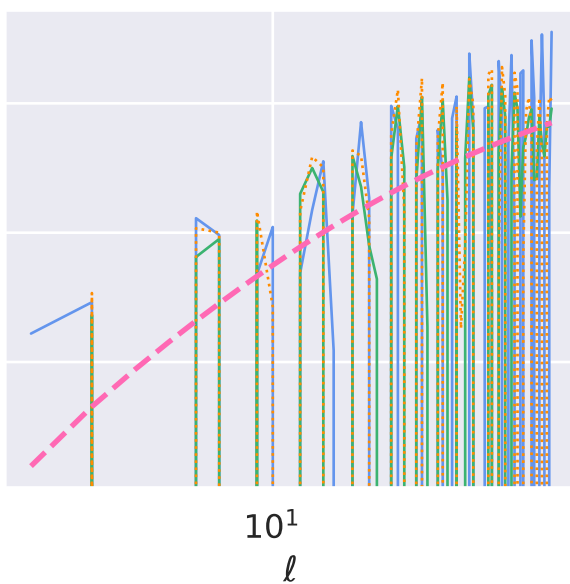
footprint (75, 105) deg



$n = 10/\text{arcmin}^2$

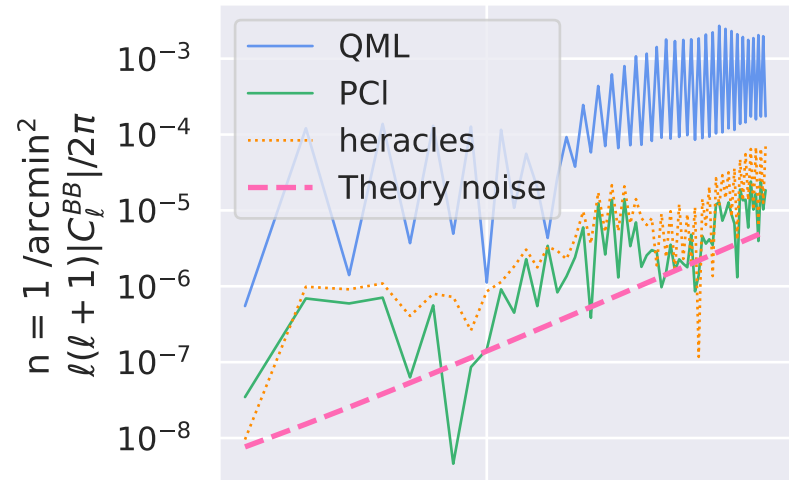


$n = 20/\text{arcmin}^2$



BB power spectrum estimates

footprint (55, 125) deg



footprint (65, 115) deg



footprint (75, 105) deg

